Instructor Guide: Lesson 2 — Data Visualization and Operational Dashboards

Duration: 1 hour

Topic-by-Topic Teaching Notes

Chart Types and When to Use Them

Open with four small data tables on the board or a shared screen: (1) ticket count by severity (three rows), (2) daily ticket count over 30 days, (3) severity-mix percentages, (4) 12 resolution codes with counts. Each table is a mini Key Performance Indicator (KPI) candidate that leadership might ask about at a 30-day review. Ask the class which chart type fits each before revealing your answer. The fourth case is the one to linger on — the reflex is to reach for a pie chart, and the fix (horizontal bar sorted by count) is the lesson's most transferable habit. If a learner says "I would use a line chart for severity" because it looks cleaner, stop and restore the discipline of "name the question first." The lesson is not memorization of chart names; it is the habit of matching visual to question.

A good pivot to land is the combo-chart case. Show the OrbitCo weekly volume vs MTTR combo chart briefly — "this one chart answers two questions at once." Contrast with the failure mode: trying to squeeze three or four unrelated metrics onto a single combo chart, which produces a visual that answers no question well. Two series on a combo chart is the practical limit.

Pivot Tables and Pivot Charts

Live-build a pivot on the OrbitCo `Tickets` Table in front of the class. Start with nothing on the rows, columns, or values, then demonstrate how dropping `Severity` into Rows and `TicketID` into Values immediately produces a count. Move `Team` into Columns — the pivot instantly becomes a crosstab. Change aggregation to "% of Row Total" — it becomes a composition view. Add a slicer on `Team` — click between teams and watch

the pivot update. The point of the demo is the speed: five questions answered in under two minutes. That speed is the argument for learning pivots instead of reaching for formulas on everything.

A common misconception to flag: the pivot does not automatically update when the source `Tickets` Table grows. Show this by adding a row to the source and proving that the pivot still shows the old count until Refresh is clicked. Then add the Refresh All shortcut to the Quick Access Toolbar and make it a habit for the rest of the course. Learners who skip Refresh publish a dashboard that silently reports last week's numbers — and the downstream Service Level Agreement (SLA) breach tile reads green when it should be red.

Building an Operational Dashboard

This is the centerpiece of the lesson and the longest activity. Before starting Activity 1 (the hands-on lab), walk the class through one complete dashboard on a shared screen — yours, not theirs — so they see the three-band layout with real content before they try to build their own. Point at the tiles: "Headlines." Point at the middle band: "Trends." Point at the bottom band: "Evidence." Then name each of the dashboard hygiene rules out loud while pointing at the evidence of it on your example. The `README` tab especially: open it, read it aloud, and say "this is what OJL 5d looks like in a spreadsheet."

During the lab, circulate specifically for three failure modes. First: pie charts with more than five slices — redirect to horizontal bar. Second: hardcoded axis labels like "Tickets" that tell the reader nothing — push for a sentence-style title ("Incidents holding steady around 14/week after the June runbook update" if the data supports it, or a neutral but specific label like "Weekly ticket volume, 30-day window" if not). Third: raw data showing on the dashboard sheet — it belongs on a separate tab every time.

Conditional Formatting as Visualization

Show two versions of the same ticket table side-by-side on the projector — one with five competing conditional-formatting rules (data bars, color scale, icon set, "breach → red fill," "high severity → bold"), one with just the "SLA breach → red fill" rule. Ask which is more useful. The five-rule version looks busier and feels more "analyzed," but

the single-rule version is the one a reader can actually use. This is the tension to name for the class: more rules does not equal more insight. The rule is there to help the reader find something, not to prove that analysis happened.

Reinforce the accessibility caveat on icon sets. A live trick: use the Windows Magnifier or a colorblind-simulator screenshot to show what a red/green-only icon set looks like to a red-green colorblind reader. The icons disappear into the background. The shape-plus-color combination (up-arrow green, down-arrow red) still reads for that same reader. A 30-second detour here pays off every time a learner builds an icon set later in the course.

Exporting Charts for Use Elsewhere

The three paste modes are worth demonstrating live, not just describing. Copy a chart from the dashboard, paste three times into a blank PowerPoint slide — once as Picture, once as Embedded Object, once with Link. Then rename the Excel file on disk and open the PowerPoint. The linked chart breaks with a `Source Not Found` dialog; the embedded object still opens but edits do not propagate; the picture is unchanged. Five minutes of demo makes the "paste linked is fragile" lesson stick in a way that a paragraph in the content cannot.

Avoiding Misleading Visualizations

Show a pair of charts made from the same OrbitCo uptime data: one with a y-axis starting at 99.5% (the drops look enormous) and one starting at 0% (the drops look like nothing). Ask: "Which is more honest? Which is more informative?" The answer is not that one is always right — it is that the chart author has to choose consciously and explain the choice. This is also the moment to preview the Lesson 4 point about annotating charts: if a non-zero baseline is necessary for readability, a plain-language annotation on the chart explaining why (e.g., "Axis starts at 99.5% to show variation in the target range") is how honest analysts handle the tradeoff.

The 3D-pie segment is quick but worth doing. Throw up a 3D pie with twelve slices drawn from real operational data, then the same data as a horizontal bar sorted by count. Ask which one they could talk about in a stand-up. The bar wins every time, and the class will not 3D-pie a chart again.

Timing Suggestions

Segment	Time	Notes
Chart Types and When to Use Them	7 min	Four-table exercise; emphasize the 12-code case
Pivot Tables and Pivot Charts	8 min	Live-build the pivot on the `Tickets` Table; demo slicer
Building an Operational Dashboard	7 min	Walk the example dashboard; name the three-band layout
Conditional Formatting as Visualization	5 min	Side-by-side rule-count comparison; accessibility detour
Exporting Charts for Use Elsewhere	4 min	Three-paste-modes live demo with a rename test
Avoiding Misleading Visualizations	4 min	Truncated y-axis pair; 3D-pie-vs-bar comparison
Activity 1: Build the OrbitCo Operational Dashboard	15 min	12 min individual work + 3 min share-out (tiles + one trend chart)
Activity 2: Chart Critique	7 min	4 min pair work + 3 min debrief
Wrap-Up and Reflection	3 min	Key takeaways; one reflection question live
Total	**60 min**	

A note on timing: the Activity 1 window in this table is the compact 60-minute delivery. For an extended lab block (90–120 minutes), expand Activity 1 to 25 minutes and add the second chart plus the conditionally-formatted details band back in. For the strict 60-minute version shown, learners build the tile strip and one trend chart; the full multi-chart dashboard is finished as Lesson 4 prep work between sessions.

Common Student Questions

Q: When should I use a pivot chart vs a regular chart fed by a pivot? A: Use a pivot chart when you expect the question to change — when the pivot will be re-sliced by team, month, or severity during a live review, the chart should follow automatically. Use a regular chart bound to a pivot's output range when the chart is stable and will be pasted into a deck; the regular chart is easier to format and less likely to break when the pivot is refactored.

Q: Can I put both the `Tickets` Table and the dashboard on the same sheet to save tabs? A: No. The dashboard sheet should be the only sheet the reader looks at; raw data belongs on separate tabs. Combining them forces the reader to scroll past hundreds of rows to find the tiles, and it makes the dashboard feel cluttered. The small cost of an extra tab is worth it every time.

Q: Why does my sparkline on the KPI tile look empty or wrong? A: Two common causes. First, the source range is empty cells or text — sparklines only render numeric data. Second, the source range shifted when you moved the tile; sparklines store an absolute source, not a structured reference, so they break when referenced data is rearranged. Delete and re-insert the sparkline against the correct range.

Q: Is it OK to use a truncated y-axis if I label it? A: It is OK in narrow cases — uptime percentages and latency rates where the useful range is concentrated near 100% (or near zero). For any other metric, a non-zero baseline on a column or bar chart is misleading and the right fix is to switch to a line chart. Always label a truncated axis explicitly with an annotation explaining why.

Q: My donut chart has three slices but one is almost invisible because it is so small. What do I do? A: Donut charts are readable only when all slices are at least about 5% of the total. If one slice is 1% and two are ~50% each, the donut answers "this is almost entirely two categories" and you should probably show the small one as a callout number instead, or label the slice with its percentage so the reader does not have to estimate from a sliver.

Facilitation Tips

- **Live-build the pivot on the projector.** The speed of pivoting is the whole argument for learning them. Showing five variations in two minutes makes the "just use formulas" resistance evaporate without having to argue about it.
- **Do not skip the three-paste-modes demo.** It is five minutes and it is the moment that turns "paste linked sounds fine" into "I understand why this breaks in practice." The rename-the-source trick is the memorable part; commit to it.
- **Protect the Chart Critique debrief.** The well-designed chart in the set is the one most learners try to find a problem with. The debrief is where you can name that reflex — "a critique exercise is not always wrong" — and reinforce the habit of defending good design, not only flagging bad.
- **Name the OJL 5d connection out loud during the dashboard section.** Say: "The `README` tab, the timestamp, the named ranges — these are the three things that make this dashboard documentation, not a disposable view. That is the competency this lesson maps to." Naming the mapping explicitly helps the class see why the hygiene rules matter beyond stylistic preference.
- **Watch the clock on Activity 1.** The 25-minute dashboard lab expands to fill whatever time is available. Give a 10-minute check-in ("tiles done, on to the trend chart") and a 5-minute warning. If a learner is still fighting the `XLOOKUP` in their KPI tile at minute 20, hand them the skeleton formula and let them finish — the lab goal is the dashboard layout and the reading habit, not formula drill.